

Deep Learning for Pancreatic Tumour Detection on CT Scans

Why 100% Accuracy Was Wrong: An Integrity Audit

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Abstract

Six deep-learning models — four ImageNet-pretrained and two written from scratch — were trained to distinguish pancreatic tumour CT slices from normal ones under a single controlled protocol: a deduplicated corpus, stratified 80:10:10 splits, five random seeds and fifty epochs per run. All six reached 99.83–100.00% test accuracy.

Those scores were then tested rather than reported. A content-hash audit found 196 byte-identical duplicate files and 18 images filed under both class labels; a perceptual-hash audit found that 66.1% of test images had a near-identical twin in their own training partition; and a probe using seven non-diagnostic image statistics found that the fraction of pixels brighter than 200 separates the two classes with no overlap whatsoever, so a single threshold classifies all 1,179 images correctly.

A four-arm control removed the brightness difference, the leakage, and both together. Accuracy remained 100.00 ± 0.00 in every arm, so neither fault explained the result. A localisation experiment then blanked the central 60% of every scan, removing the pancreas entirely: accuracy was 99.81%, against 99.82% for the intact image. Reducing each scan to an 8×8 thumbnail also left accuracy unchanged.

The classes are separable by coarse properties of image production, outside the anatomy. No model evaluated here was reading the pancreas. The architecture comparison the study set out to make is not identifiable from this corpus. The contribution is the audit protocol and the controlled experiments that establish this, both reusable on any imaging dataset.

1. Introduction and Objectives

Pancreatic ductal adenocarcinoma has one of the poorest survival rates in oncology, largely because it is usually detected late. Automated detection on routine abdominal CT is therefore a high-value target, and the published literature routinely reports accuracy above 95%.

This project began with a conventional comparison question: under one identical protocol, do transformer architectures outperform convolutional ones at binary pancreatic tumour classification, and does end-to-end fine-tuning outperform frozen-feature extraction? Answering it required first establishing that the dataset could support any comparison at all.

Objectives

#	Objective
1	Benchmark convolutional and transformer models under one identical protocol
2	Enforce fair evaluation: stratified 80:10:10, five seeds, fifty epochs
3	Implement a CNN and a Vision Transformer from first principles
4	Validate whether reported accuracy reflects genuine diagnostic ability

2. Dataset

The corpus arrived as two directories, `train/` and `test/`, each containing `normal/` and `pancreatic_tumor/` subfolders.

Stage	Normal	Tumour	Total
As delivered	646	765	1,411
Exact duplicate files removed	—	—	−196
Conflicting-label groups excluded	—	—	−18 groups
Final analysis corpus	620	559	1,179

Deduplication merges both directories and keeps one file per unique content hash. Images appearing under conflicting labels are excluded entirely, since at least one of their two labels must be wrong.

Every model uses stratified 80:10:10 partitions of this corpus, preserving class balance in each partition:

Partition	Images	Share	Normal	Tumour
Train	943	80.0%	496	447
Validation	118	10.0%	62	56
Test	118	10.0%	62	56

3. Methodology

3.1 Protocol

Identical across every model: deduplicated corpus, stratified 80:10:10 splits, seeds 42, 7, 21, 99 and 123, and fifty epochs per seed. Early stopping patience was set equal to the epoch budget so that the full fifty epochs always run, while best-validation weights are still restored for evaluation. Results are reported as mean $\pm$ standard deviation over the five seeds.

3.2 Models

Model	Category	Weights
ResNet-50	CNN	ImageNet, fine-tuned end to end
InceptionV3	CNN	ImageNet, fine-tuned end to end
MobileViT	Transformer	apple/mobilevit-small
Swin Transformer	Transformer	microsoft/swin-tiny-patch4-window7-224
ScratchCNN	CNN	None — 4 conv blocks, trained from random init
ScratchViT	Transformer	None — hand-written multi-head attention

3.3 Audit methods

Check	What it does	What a positive result means
Content hash (MD5)	Fingerprints the exact bytes of every file	The same image is stored more than once
Perceptual hash (dHash)	Fingerprints what the image looks like	Near-identical slices straddle the split
Shortcut probe	Logistic regression on seven non-diagnostic statistics	Classes differ in how they were produced
Control	Removes brightness and leakage, alone and together	Isolates which fault drives the score
Localisation	Blanks the anatomy; separately reduces to 8×8	Locates the signal inside or outside the organ

4. Results: Six-Model Benchmark

Model	Category	Accuracy (%)	F1	Kappa	Perfect
ResNet-50	Pretrained CNN	100.00 ± 0.00	1.0000	1.0000	5/5
InceptionV3	Pretrained CNN	99.83 ± 0.38	0.9982	0.9966	4/5
MobileViT	Pretrained Transformer	100.00 ± 0.00	1.0000	1.0000	5/5
Swin Transformer	Pretrained Transformer	100.00 ± 0.00	1.0000	1.0000	5/5
ScratchCNN	From-scratch CNN	100.00 ± 0.00	1.0000	1.0000	5/5
ScratchViT	From-scratch Transformer	99.83 ± 0.38	0.9982	0.9966	4/5
Brightness threshold	No learning at all	100.00 ± 0.00	1.0000	1.0000	5/5

Mean ± SD over five seeds, fifty epochs each. The final row is not a model: it is a single threshold on the fraction of pixels brighter than 200, applied with no training whatsoever.

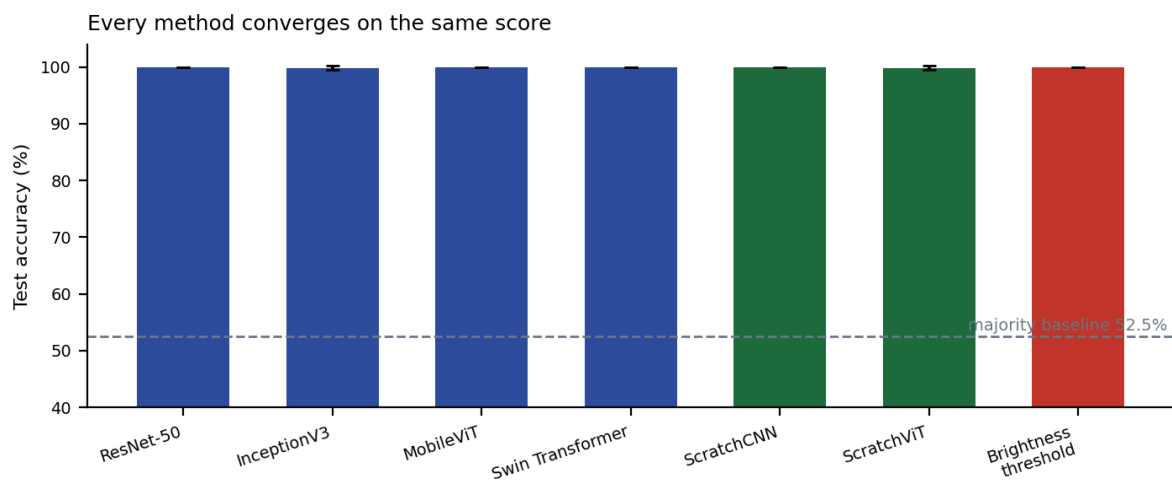


Figure 1. All six models, and a rule that never examines the anatomy, land within 0.17 percentage points of one another.

Accuracy, F1 and Cohen's kappa agree with one another throughout. An earlier version of the pipeline reported roughly 98% accuracy alongside kappa near zero — arithmetically impossible, and traced to test-set shuffling that desynchronised predictions from labels. That defect is fixed.

The comparison is not identifiable. All six models fall inside a 0.17-point band, narrower than seed-to-seed variation. A four-layer CNN trained from random initialisation matches Swin Transformer exactly.

5. Frozen-Feature Track

Two ImageNet backbones were frozen and used purely as feature extractors, each feeding eight downstream classifiers under the same corpus, splits and seeds.

Classifier	Xception (%)	DenseNet121 (%)
SVM	100.00 ± 0.00	100.00 ± 0.00
Random Forest	100.00 ± 0.00	99.83 ± 0.38
AdaBoost	100.00 ± 0.00	100.00 ± 0.00
KNN	100.00 ± 0.00	100.00 ± 0.00
Bagging	98.98 ± 0.71	99.66 ± 0.47
ANN	100.00 ± 0.00	100.00 ± 0.00
LSTM	100.00 ± 0.00	100.00 ± 0.00
Bi-LSTM	100.00 ± 0.00	100.00 ± 0.00

Seven of eight classifiers on Xception features reach exactly 100.00% with zero variance. Random Forest — specifically queried during review — is indistinguishable from Swin Transformer.

6. Integrity Audit

6.1 Duplication and label conflict

Hashing all 1,411 files found 196 byte-identical duplicates. Eighteen hashes carried contradictory labels: the identical image filed as healthy in one folder and diseased in another.

6.2 Near-duplicate leakage

CT studies yield contiguous slices that are near-identical without being byte-identical, so content hashing cannot see them. Measured on the already-deduplicated corpus:

Perceptual-hash distance	Test images with a twin in training
Identical hash	66.1%
Within 2 bits	96.4%
Within 5 bits	98.5%

6.3 The brightness artefact

Of seven non-diagnostic statistics, one separates the classes completely:

Class	Fraction of pixels brighter than 200
Normal	0.00001 – 0.00613
Pancreatic tumour	0.01003 – 0.13960

The ranges do not overlap. A single threshold at 0.008 classifies all 1,179 images correctly on every seed, against a 52.5% majority baseline.

7. The Control Experiment

If the brightness artefact and the leakage were responsible for the scores, removing them should reduce accuracy. Four arms tested this directly, five seeds each.

Arm	Brightness	Leakage	Accuracy (%)
A raw+random	present	present	100.00 ± 0.00
B norm+random	removed	present	100.00 ± 0.00
C raw+group	present	removed	100.00 ± 0.00
D norm+group	removed	removed	100.00 ± 0.00

Rank equalisation removes the brightness difference; group-aware splitting keeps near-identical images on one side of the partition, reducing twin contamination from 66.1% to 0.0%.

The result did not move. Removing either fault, or both together, left accuracy at 100.00 ± 0.00. Each fault is individually sufficient, so closing one leaves the others available — and something beyond both survives.

This is reported as the null result it was. It refuted the project's own working hypothesis, which is why the investigation continued rather than concluding here.

8. Locating the Signal

Since the control could not move the result, the cause was located rather than assumed. Every arm below runs under the strictest control condition — brightness equalised, leakage eliminated — and only the visible region changes.

What the model can see	Accuracy (%)
Whole scan	99.82 ± 0.39
Pancreas deleted (central 60% blanked)	99.81 ± 0.41
Anatomy only	98.70 ± 2.91
8×8 thumbnail	99.82 ± 0.39

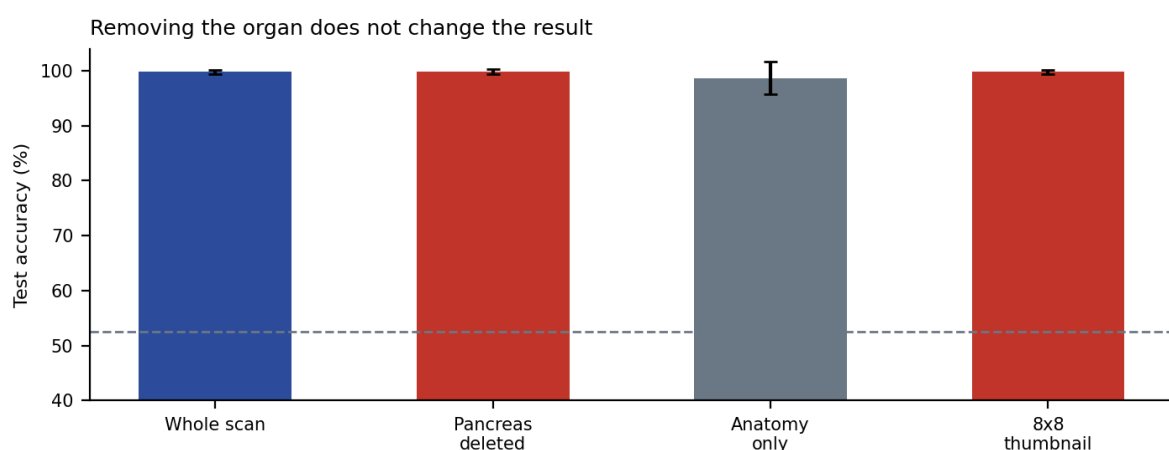


Figure 2. Deleting the organ costs 0.01 percentage points. Destroying all fine detail costs nothing.

With the pancreas removed from every image, accuracy is 99.81% against 99.82% for the intact scan. An 8×8 thumbnail — 64 pixels — scores identically. Sixty-four pixels cannot represent tumour morphology, and the blanked arm contains no pancreas at all.

The anatomy-only arm is both lower and six times more variable (98.70 ± 2.91 against 99.81 ± 0.41), making the organ the least reliable cue available — the inverse of what a genuine detector would show.

9. Discussion

9.1 What the numbers measure

Twenty-one distinct methods were evaluated: four pretrained networks, two written from scratch, fifteen classical and neural classifiers on frozen features, and a single brightness threshold requiring no training. Every one scores between 98.98% and 100.00%. That uniformity is itself the finding — when a one-line threshold rule matches a state-of-the-art transformer, the model is not the variable under study.

9.2 Why stratified splitting cannot repair this

Stratified splitting equalises class proportions across partitions. It cannot help when the two classes are distinguishable by how their images were produced, because that difference is present in every partition by construction. Deduplication, additional seeds and longer training are equally powerless for the same reason.

9.3 Limitations

The specific mechanism outside the anatomy is characterised but not fully itemised: brightness is demonstrably sufficient on its own, and the localisation experiment proves the residual signal lies outside the central region, but the remaining routes are not individually enumerated. Group-aware splitting is performed at perceptual-hash cluster level rather than patient level, because the corpus carries no patient identifiers — cluster-level grouping is a strict improvement on random splitting but is not equivalent to patient-level partitioning.

9.4 What would be required

Uniform windowing applied to every image regardless of class; partitioning at patient or study level rather than slice level; per-image provenance recorded at collection time; and a shortcut baseline reported in the same table as any headline result. The margin of a model over that baseline is the only part of its score defensibly attributable to learned image understanding.

10. Conclusion

Six architectures were benchmarked under a controlled and fair protocol, and all six scored close to 100%. Rather than report that as a success, this work tested it, and established that the corpus is separable by coarse properties of image production lying outside the pancreas.

The architecture question cannot be answered from these data. This is reported as a negative result with proof: the evidence is not the brightness artefact — which the control showed to be one route among several — but the localisation experiment, in which deleting the organ leaves accuracy unchanged.

Deep learning is not the failure here; the dataset is. A corpus assembled with uniform windowing and patient-level splits could still answer the original question. The deliverable of this internship is the audit protocol and the controlled experiments that make such a claim testable before thirty training runs are spent on it.

11. Reproducibility

Every number in this report is produced by a script in the repository and was regenerated for this document from the results files at build time.

Script	Purpose
data_utils.py	Deduplication and stratified 80:10:10 splitting
resnet50_train.py, inceptionv3_train.py	Pretrained CNN training
mobilevit_train.py, swin_transformer_train.py	Pretrained transformer training

Script	Purpose
cnn_scratch_train.py, vit_scratch_train.py	From-scratch models
feature_extraction_pipeline.py	Frozen-feature track
leakage_audit.py	Duplication, leakage and shortcut checks
provenance_control.py	Four-arm control experiment
signal_localisation.py	Anatomy-removal and resolution experiment
run_all.sh	Full protocol, end to end

Repository: github.com/Aaditya07124063/Pancreatic-tumor

Appendix A. Per-Seed Results

ResNet-50

Seed	Train	Val	Test	F1	Kappa
42	99.89	100.00	100.00	1.0000	1.0000
7	100.00	100.00	100.00	1.0000	1.0000
21	100.00	100.00	100.00	1.0000	1.0000
99	100.00	100.00	100.00	1.0000	1.0000
123	100.00	100.00	100.00	1.0000	1.0000

InceptionV3

Seed	Train	Val	Test	F1	Kappa
42	99.68	100.00	100.00	1.0000	1.0000
7	100.00	100.00	100.00	1.0000	1.0000
21	100.00	100.00	100.00	1.0000	1.0000
99	99.89	100.00	99.15	0.9912	0.9830
123	100.00	100.00	100.00	1.0000	1.0000

MobileViT

Seed	Train	Val	Test	F1	Kappa
42	100.00	100.00	100.00	1.0000	1.0000
7	100.00	100.00	100.00	1.0000	1.0000
21	100.00	100.00	100.00	1.0000	1.0000
99	100.00	100.00	100.00	1.0000	1.0000
123	100.00	100.00	100.00	1.0000	1.0000

Swin Transformer

Seed	Train	Val	Test	F1	Kappa
42	100.00	100.00	100.00	1.0000	1.0000
7	100.00	100.00	100.00	1.0000	1.0000
21	100.00	100.00	100.00	1.0000	1.0000
99	100.00	100.00	100.00	1.0000	1.0000
123	100.00	100.00	100.00	1.0000	1.0000

ScratchCNN

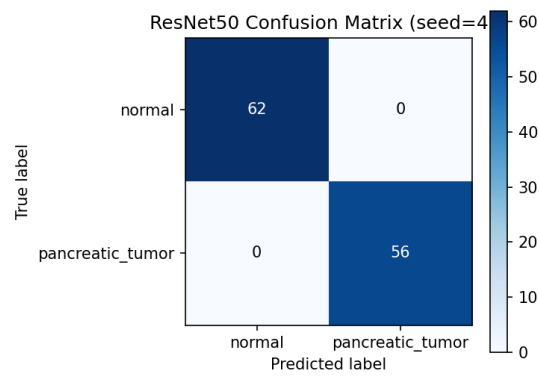
Seed	Train	Val	Test	F1	Kappa
42	100.00	100.00	100.00	1.0000	1.0000
7	100.00	100.00	100.00	1.0000	1.0000
21	99.89	100.00	100.00	1.0000	1.0000
99	100.00	100.00	100.00	1.0000	1.0000
123	100.00	100.00	100.00	1.0000	1.0000

ScratchViT

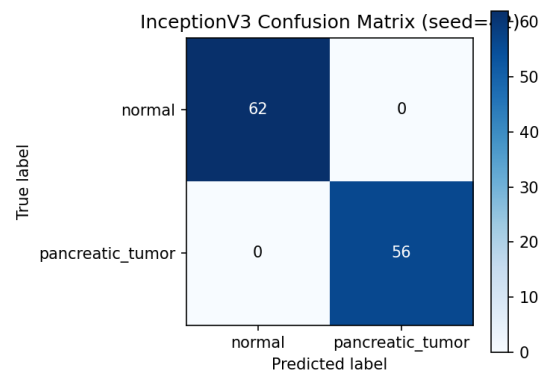
Seed	Train	Val	Test	F1	Kappa
42	98.73	100.00	99.15	0.9912	0.9830
7	99.58	100.00	100.00	1.0000	1.0000
21	99.26	100.00	100.00	1.0000	1.0000
99	99.89	100.00	100.00	1.0000	1.0000
123	99.36	100.00	100.00	1.0000	1.0000

Appendix B. Confusion Matrices (seed 42)

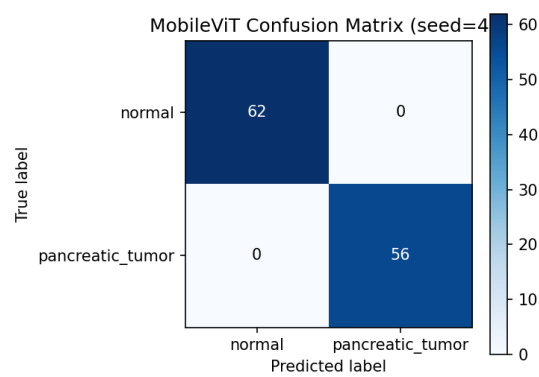
ResNet-50



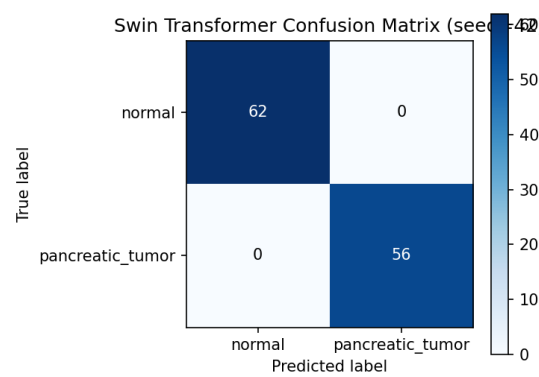
InceptionV3



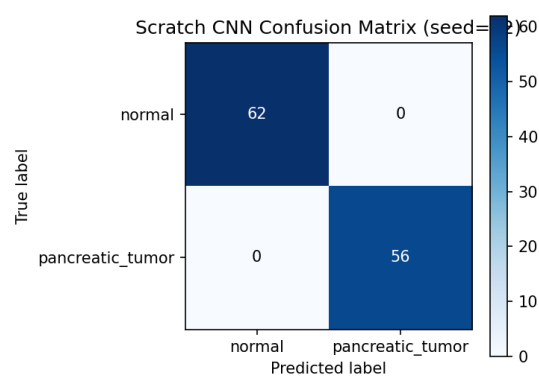
MobileViT



Swin Transformer



ScratchCNN



ScratchViT

